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19 January 1978
SUPERSEDING
MIL-R-81493(AS)
23 January 1968

MILITARY SPECIFICATION

RADIO SET AN/PRC-90

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 The equipment covered by this specification shall provide an emergency beacon tone transmission at 243.0 megahertz (MHz), a keyed tone modulated continuous wave (mcw) 243.0 MHz and provide two-way AM voice communication at 243.0 MHz and 282.8 MHz. Simultaneous operation at 243.0 MHz and 282.8 MHz is not required.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-5272	Environmental Testing, Aeronautical And Associated Equipment, General Specification For
MIL-E-5400	Electronic Equipment, Airborne, General Specification For
MIL-N-7513	Nomenclature Assignment, Contractors Method For Obtaining
MIL-E-17555	Electronic And Electrical Equipment Accessories, And Repair Parts: Packaging And Packing Of
MIL-T-18303	Test Procedures; Preproduction, Acceptance, And Life, For Aircraft Electronic Equipment, Format For

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5820

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STANDARDS

Military

MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-480	Configuration Control-Engineering Changes, Deviations And Waivers
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-794	Parts And Equipment, Procedures For Packaging And Packing Of

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction (First Article). This specification makes provision for preproduction (First Article) testing.

* 3.2 Parts and materials. In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

a. The circuit modules shall be considered nonreparable subassemblies.

b. Other parts and materials requirements shall conform to MIL-E-5400.

c. Nonreparable subassemblies, as outlined in MIL-E-5400, shall be used when practicable. The general size of the subassembly and the amount of circuitry to be included therein shall be approved by the procuring activity. Nonreparable subassemblies must be reliable. (See 6.4)

d. When previously produced models of this equipment did not use nonreparable subassemblies, the design shall not be changed to employ nonreparable subassemblies without the approval of the procuring activity.

3.2.1 Nonstandard parts and materials approval. Approval for the use of nonstandard parts and materials shall be obtained as outlined in MIL-E-5400.

* 3.2.2 Microelectronic modular assemblies. The circuit modules shall be considered to be nonreparable subassemblies; however, their circuit and component layouts, parts lists, schematic diagrams and test procedures shall be included as part of the First Article Test Report. After First Article approval, only Class I and Class II changes in accordance with MIL-STD-480 shall be made, and all such changes must be submitted to the procuring activity for approval. This includes all module parts changes other than for standard parts.

3.3 Design and construction. The equipment shall conform with all the applicable requirements of MIL-E-5400 for design, construction and workmanship, except as otherwise specified herein.

3.3.1 Total weight. The total weight of the equipment shall be a minimum consistent with good design and shall not exceed 1.6 pounds.

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3.3.2 Reliability.

3.3.2.1 Service life. The design of the equipment shall be such, that with reasonable handling and replacement of parts, a useful service life of five years can be expected. Parts requiring scheduled replacement shall be specified by the contractor.

* 3.3.2.2 Reliability in mean time between failures (mtbf). For the test program under 4.4.3, the specified mean (operating) time between failures shall be 2000 hours.

* 3.3.3 Battery type. The equipment shall be designed to operate with a BA-1568/U battery.

3.3.4 Interchangeability of reordered equipment. For reordered equipment interchangeability shall exist between all replaceable assemblies, sub-assemblies, and parts of a designated model of any previously manufactured equipment supplied or designated by the procuring activity.

3.3.5 Maintenance provisions and field testing. Provisions for maintenance shall be as specified in MIL-E-5400. Specific test points shall be provided to the maximum extent practicable consistent with the other requirements of this specification. The design of the equipment shall be such as to allow antenna replacement and routine "GO-NO-GO" testing by relatively untrained personnel using special support equipment.

* 3.3.6 Nomenclature and nameplates. Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with MIL-N-7513.

* 3.3.7 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room ambient (25° Celsius (C) ±5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90 percent relative humidity
Input power voltage	12 volts direct current (DC) (+0, -0.5 volts)
Antenna termination	50 ±j0 ohms

* 3.3.8 Service conditions. The equipment shall perform as specified herein under any combination of the following service conditions:

Temperature	-30°C to +50°C
Altitude	0 to 40,000 feet (30 to 5.5 inches Hg)
Humidity	0 to 100 percent
Input Voltage	9.5 to 14.0 volts DC

3.3.8.1 Vibration. The equipment less battery shall provide specified performance after being subjected to the vibration requirement of Curve IV of MIL-E-5400.

3.3.8.2 Temperature. The equipment shall provide specified performance over the service temperature range of -30°C to +50°C and shall withstand storage over the temperature range of -60°C to +60°C.

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* 3.3.8.3 Altitude. The equipment shall provide specified performance in voice transmit and receive modes to altitude of 10,000 feet, and in the beacon mode to 40,000 feet; except that the beacon side tone shall not be required to operate at altitudes greater than 10,000 feet. The equipment housing and seals shall withstand, without damage, the forces due to explosive decompression associated with a pressure differential of sea level (29.92 inches of Mercury (Hg)) to 40,000 feet (5.54 inches Hg).

* 3.3.8.3.1 Altitude equalization. The equipment shall meet all requirements contained herein within five minutes after sudden exposure to a difference between internal and external pressure equivalent to a change in altitude from 3000 meters (9843 feet) and sea level.

* 3.3.8.4 Sand and dust. The equipment shall provide specified performance after being covered to a depth of two inches with sand (140-mesh silica flour).

* 3.3.8.5 Fungus. The equipment shall conform to Requirement 4, Fungus Inert Materials, of MIL-STD-454.

* 3.3.9 Battery limit conditions. The equipment shall be designed to provide specified performance over the range of battery voltage from the high limit of 14 volts to the low limit of 9.5 volts.

3.3.10 Warm-up time. The time required for the equipment to warm-up prior to operation shall be kept to a minimum and shall not exceed one second under standard conditions and 5 seconds under extreme service conditions.

3.3.11 Input power requirements. The input power requirements of the equipment shall be a minimum consistent with the other requirements of this specification.

* 3.3.11.1 Input current requirements. Under standard conditions, the input current shall not exceed the following:

a.	Transmit voice: (No modulation)	100 milliamperes (ma)
b.	Transmit beacon:	70 ma
c.	Transmit MCW:	70 ma
d.	Receive: (No input signal)	40 ma

* 3.3.11.2 Overload protection. The transmitter shall not be damaged under any variation of output loading, such as grounding, partial grounding or disconnection of the antenna during transmitter operation. An input signal of 0.5 volts rms measured at the antenna connector shall not damage or prevent operation of the receiver. Minimum test time for each of the above conditions shall be five minutes.

3.3.11.3 Undervoltage protection. The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

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3.3.11.4 Voltage regulation. Provision shall be made for incorporation of voltage regulation circuitry which might be required in the event of change of battery type.

* 3.3.11.5 Overvoltage protection. The equipment shall not be damaged and its operation shall not be impaired or prevented by application of any supply voltage up to 15 volts.

3.4 Performance. Unless otherwise specified, values set forth to establish the requirements for satisfactory performance apply to performance under standard, service and battery limit conditions. When reduced performance under service or battery limit conditions is acceptable, tolerances or values setting forth acceptable variations from the performance required under standard conditions are specified under detailed requirements.

* 3.4.1 Frequency characteristics. The equipment shall provide voice, beacon, and mcw communications on one fixed channel of 243.0 MHz, and voice communications on a second fixed channel of 282.8 MHz. The design and construction of the equipment shall provide for switch selection of function and channel. The design shall allow alternate frequency choices in the range 242 to 245 MHz and in the range 281 to 284 MHz by installation of appropriate crystals.

3.4.2 Transmitter characteristics.

* 3.4.2.1 Transmitter power output.

* 3.4.2.1.1 Standard conditions. Under standard conditions, the peak envelope power (pep) output shall be a minimum of 500 milliwatts (mW) in beacon or mcw mode and 400 mW in voice mode with 80 percent modulation at 1000 Hz.

* 3.4.2.1.2 Test set conditions. With an input voltage of 10.25 volts (+0 -.25 volts) DC, the average (rms) power output shall be a minimum of 80 mW in beacon or cw mode and 40 mW in voice mode unmodulated. These output levels are to be measured by means of an approved calorimetric milliwattmeter or other approved test set.

3.4.2.1.3 Service conditions. Under any reasonable combination of service conditions, pep output shall be a minimum of 250 mW in beacon or mcw mode, and 200 mW in voice mode with 80 percent modulation at 1000 Hz.

* 3.4.2.2 Transmitter frequency stability. The frequency of the transmitter shall not deviate more than ±0.005 percent from the assigned frequency of either channel under service conditions.

3.4.2.3 Voice modulation characteristics.

3.4.2.3.1 Fidelity. For sinewave input signals over the range of 500 to 3000 Hz the voltage response of the amplification and modulation circuits shall be within +1 decibel (dB), -3 dB of the response at 1000 Hz.

3.4.2.3.2 Modulation capability. A minimum of 80 percent modulation shall be attained for a sine wave input signal equivalent to that produced by a sound pressure level of 115 dB (referred to 0.0002 dynes per square centimeter) at 1000 Hz at the face of the microphone.

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* 3.4.2.3.3 Distortion. Under standard conditions, for sinewave modulation the total harmonic distortion of the audio envelope shall not be greater than 10 percent for the input level required to produce 80 percent (+10, -0 percent) symmetrical modulation at 1000 Hz. Under service or low battery limit conditions the audio distortion shall not be greater than 15 percent for the input required to produce 80 percent modulation at 1000 Hz.

* 3.4.2.3.4 Audio input. For purposes of measuring fidelity, modulation capability, and distortion, the input shall be that produced by the microphone at the specified audio pressure level.

3.4.2.4 Tone modulation.

* 3.4.2.4.1 Beacon modulation. When operated in the beacon mode, the carrier shall be amplitude modulated (am) with a rectangular wave, periodically varying audio frequency tone. The rectangular wave modulation shall provide an on/off duty cycle with an average on time duration at the specified output power level of from 20 to 30 percent of the cycle. The rectangular wave, audio frequency tone shall periodically sweep from 1000 Hz +20, -0 percent to 300 Hz ±20 percent at a rate of from 2 to 3 sweeps per second. The power output level during the off-time shall not exceed 15 percent of the power output level during the on time. The audio tone retrace from 300 Hz to 1000 Hz shall not exceed 5 percent of the total audio tone sweep cycle time.

* 3.4.2.4.2 MCW modulation. When operated in the mcw mode, the carrier shall be am during the keyed-on period with a rectangular wave of 1000 Hz ±20 percent. The rectangular wave modulation shall conform in duty cycle, pep, and modulation percentage to the requirements of 3.4.2.1 and 3.4.2.4.1.

3.4.2.4.3 Side tone. An audible indication of beacon or mcw tone operation shall be provided to the speaker or earphone connection during these modes of operation in accordance with 3.4.3.3.2.

* 3.4.2.5 Radio frequency (rf) termination. The rf portions of the equipment shall be terminated by the flexible whip antenna. For power output and current measurement, a 50 ohm resistive load shall be substituted for the antenna.

3.4.3 Receiver characteristics.

* 3.4.3.1 Receiver sensitivity. Under standard conditions and low battery voltage (9.5 volts) with an input signal of 10 "hard" microvolts, modulated 30 percent at 1000 Hz applied to the receiver, the audio output voltage shall not be less than 6 volts peak-to-peak with a signal plus noise to noise ratio not less than 6 dB. One "hard" microvolt is defined as that rf signal at the input of a receiver when the receiver is connected through a 50 ohm 6 dB attenuator to a signal generator with its output set at 1 microvolt, thus 10 "hard" microvolts is the RF signal at the input of the receiver when connected through a 50 ohm 6 dB attenuator to a signal generator set for 10 microvolts output. Under other (service) conditions the sensitivity shall not vary from that required under low voltage and standard conditions by more than 6 dB.

* 3.4.3.2 Selectivity. The overall selectivity of the receiver shall be such that for input signals differing from the selected center frequency by ±50 kHz, the audio output voltage shall not vary by more than 6 db from that voltage produced by a center frequency input signal of equal amplitude. The receiver shall exhibit at least 55 dB signal rejection (compared to center frequency response) at all frequencies ±10 mHz or more from center frequency.

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3.4.3.3 Receiver audio characteristics. The audio characteristics of the receiver shall be as set forth below.

* 3.4.3.3.1 Audio output level. Under standard conditions or low battery voltage (9.5 volts), the receiver shall be capable of delivering an audio output of at least 6.0 volts peak to peak (p-p) into 4,000 ohm resistive load when a signal level between 10 hard microvolts and 10,000 hard microvolts, modulated 30 percent at 1000 Hz, is applied to the receiver input terminals. At 50,000 hard microvolts input (overload) the audio output shall not decrease more than 10 dB. Under service conditions the output shall not be more than 6 db below that required under standard conditions.

* 3.4.3.3.2 Volume control. The receiver audio output of the speaker and earphone terminals shall be continuously and smoothly adjustable from a barely discernable level to maximum by means of a potentiometer. Volume control variation shall not affect the power output nor degree of modulation in the transmit modes. The beacon and mcw sidetone shall be at a fixed level of between 0.2 and 1.0 volts p-p at the 4000 ohm earphone or speaker terminals, independent of the volume control setting.

3.4.3.3.3 Fidelity. For modulation frequencies from 500 Hz to 3000 Hz, the receiver output voltage shall not vary by more than +2 dB -5 dB relative to the output voltage at 1000 Hz. For modulation frequencies outside the 500 to 3000 Hz range the output shall be attenuated to the maximum extent practicable without recourse to inductive filter components.

* 3.4.3.3.4 Distortion. Under standard conditions, with an input signal of 100 "hard" microvolts modulated 30 percent at 1000 Hz and with the volume control set to provide 6 volts p-p audio output, the total harmonic distortion content of the receiver audio output shall not exceed 10 percent. Under services or low limit conditions the distortion shall not exceed 15 percent.

3.4.3.3.5 Audio termination. For measurement of audio output fidelity, and distortion, the receiver shall be terminated in a 4,000 ohm resistive load.

* 3.4.3.4 Receiver frequency stability. Under any reasonable combination of service conditions and battery limits specified herein, the frequency stability of the receiver shall be such as to permit the reception of an input signal of the specified intensity anywhere within 0.005 percent of either assigned frequency of 243.0 or 282.8 MHz.

3.4.3.5 Receiver radiation. The receiver shall be nonradiating to the extent that two radio sets may be operated in the receive mode within 10 feet of one another without loss of sensitivity of more than 1 dB under any combination of channel selections.

3.5 Detailed requirements.

3.5.1 Functions. The transmitter portion of the radio shall radiate AM rf power from the flexible whip antenna on either assigned carrier frequency of 243.0 or 282.8 MHz. The operating frequency shall be selected by a switch. The receiver portion of the radio set shall receive AM rf energy at the same frequency as the selected transmitter channel, from the same flexible whip antenna, and convert it to aural intelligence through a built-in speaker or external earpiece. The transmitter shall provide three manually selected modes of operation: (1) voice operation on either channel using a built-in microphone; (2) manually keyed mcw operation consisting of 243 MHz rf carrier

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amplitude-modulated by a 1 kHz square wave tone; (3) a beacon signal consisting of a 243.0 MHz carrier amplitude modulated by a distinctive square-wave swept tone to identify the transmission as a distress beacon signal. Automatic beacon actuation is not required. When operated in the mcw or beacon mode, an audible indication of equipment operation shall be provided through the speaker or external earpiece. The equipment shall be self-contained including flexible antenna, switches, microphone, speaker, earphone connection, and battery.

* 3.5.2 Form factor. The radio set shall be rectangular in shape, with maximum dimensions of 6.0 inches high, 3.1 inches wide, 1.4 inches deep excluding the wrist strap, antenna and earphone connection. There shall be no exposed sharp corners or edges. The unit shall be capable of operation when held in either hand. When the equipment is in use, the longest dimension shall be in the vertical plane. The speaker and microphone shall be flush mounted in one side of the housing. The flexible antenna shall be attached to the upper part of the housing. Antenna shall be removable for replacement and testing. The battery cavity shall be designed such that the battery can only be inserted with proper polarity. An easily removable cover, of the captive type, shall be provided to accomplish sealing of the battery cavity and to permit periodic battery replacement. The design and construction of the housing and the battery cavity shall provide a protective seal such that the electronic portions of the equipment are not exposed to the entrance of moisture, contaminants or foreign objects when the battery cover is removed. A wrist strap shall be provided as part of the equipment Provisions shall be made for holding the antenna in the stowed position.

* 3.5.3 Contents. The primary electronic functions of the radio set shall be performed by film hybrid microcircuits. The radio frequency portions of the equipment shall be independent for each channel, that is, the receiver and transmitter for 243.0 MHz shall be separate from those used for 282.8 MHz. The total radio set including antenna, manual switches, speaker, microphone, earphone connection, battery and electronic assembly shall be waterproof, dustproof, and shall be capable of immersion in salt water to a depth of 2 feet for a period of 24 hours, and a depth of 50 feet for 5 minutes, without mechanical or electrical damage.

3.5.3.1 Speaker and microphone. Separate microphone and speaker shall be provided. The response under standard conditions shall comply with the following when tested in accordance with American Standards Association (ASA) procedures:

a. The microphone sensitivity shall be -75 ±3 dB relative to 1.0 volt per microbar at 1000 Hz when connected to a 4000 ohm resistive load.

b. The speaker sensitivity (output) shall be 100 ±3 dB sound pressure relative to 0.0002 microbar at 1000 Hz when energized by 0.1 milliwatt 4,000 ohm source (1.27 volts in series with a 4,000 ohm resistor) and terminated by a Type I coupler.

* 3.5.3.2 Antenna and RF impedance.

* 3.5.3.2.1 Impedance. The RF output impedance at the antenna feed point shall be 50 ohms resistive at the frequency to which the radio is switched

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* 3.5.3.2.2 Antenna. A half-wavelength end-fed vertical monopole antenna shall be supplied as part of the radio set. Construction shall be flexible, telescopic, with a matching network at its base, and it shall be stowable by telescoping and bending down along the side of the radio set. The antenna shall be impedance matched to the 50 ohm antenna feed point of the radio set with a voltage standing wave ratio (vswr) of 1.5 to 1 or less at 243.0 and 282.8 MHz.

3.5.3.3 Manual controls. The manual controls of the radio set shall consist of:

a. A rotating selector knob which selects channel and mode of operation.

b. A push button which energizes the transmitter in voice operation.

c. A push button which energizes the transmitter in mcw operation.

d. A volume control.

e. An earphone connection.

* 3.5.3.3.1 Mode selector. The mode selector knob shall be recessed in the radio set and shall perform mode and channel switching functions. The mode selector shall provide four positions. 282.8/VOICE, OFF, 243/VOICE/MCW, and 243/BCN. Each position shall be identified by a positive detent and by a block lettered label. In the 282.8/VOICE position, the receiver and transmitter shall be tuned to 282.8 MHz. In the OFF position, the radio set shall be de-energized. In the 243/VOICE/MCW position, the receiver and transmitter shall be tuned to 243.0 MHz. In the 243 BCN position, the receiver shall be de-energized and the transmitter shall continuously radiate the beacon on 243.0 MHz.

3.5.3.3.2 Push to talk button. When the mode switch is on 282.8/VOICE or 243/VOICE/MCW, depressing the push to talk push-button shall de-energize the receiver and activate the transmitter in voice transmission mode, giving push-to-talk operation on the channel selected by the mode switch.

3.5.3.3.3 MCW push button. When the mode switch is on 243/VOICE/MCW, depressing the MCW push-button shall de-energize the receiver and shall cause the transmitter to radiate an MCW tone on 243.0 MHz giving keyed MCW break-in operation. When the mode switch is on any other position, depressing the MCW push-button shall cause no effect.

* 3.5.3.3.4 Volume control. The volume control shall be operative in receive mode and shall be marked with an arrow indicating the direction of rotation which will increase volume. The word MAX will appear at the head of the arrow.

3.5.3.3.5 Earphone connection. Engaging the earphone connection shall automatically turn off the internal speaker. Damage to the internal circuits shall not occur when the earphone connection is terminated in any load from short to open circuit. When the earphone connection is not engaged, shorting the earphone connection shall not cause any change in equipment operation. The connection shall be identified by an adjacent label on the external face of the radio set housing.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections ar deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of tests. Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements:

a. Preproduction qualification (First Article) tests

b. Initial production tests

c. Production acceptance tests

d. Life tests

4.2 Preproduction (First Article) tests. Preproduction (First Article) tests shall be made on equipment representative of the production equipments to be supplied under the contract except that operating frequencies shall be 242.5 and 282.8 MHz. Preproduction (First Article) tests shall be the responsibility of the contractor and shall be conducted in accordance with the approved tests procedures of 4.5. The Government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government.

4.2.1 Preproduction (First Article) test data. The contractor shall submit all data collected in conducting these tests to the procuring activity for review and approval. On first production contracts these data shall include a list of all electrical and electronic parts giving their specified voltage, current and temperature ratings and the applied circuit voltage, current and ambient and surface temperature shall be obtained under the extreme high temperature operating condition.

* 4.2.2 Scope of tests. Preproduction (First Article) tests shall be in accordance with the approved test procedure submitted in accordance with 4.5 and shall include the following tests and all additional tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of the specification and the contract.

Room Ambient up to 90% Relative

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Performance Tests

Parameter	Paragraph Reference
Transmitter power output	3.4.2.1
Beacon modulation	3.4.2.4.1
MCW modulation	3.4.2.4.2
Fidelity	3.4.2.3.1
Modulation capability	3.4.2.3.2
Distortion	3.4.2.3.3
Side tone	3.4.2.4.3 and 3.4.3.3.2
Transmitter frequency stability	3.4.2.2
Receiver sensitivity	3.4.3.1
Selectivity	3.4 3.2
Audio output level	3.4.3.3.1
Volume control	3.4.3.3.2
Fidelity	3.4.3.3.3
Distortion	3.4.3.3.4
Receiver frequency stability	3.4.3.4

Design Requirements Tests

Parameter	Paragraph Reference
Total weight	3.3.1
Warm-Up time	3.3.10
Input current requirements	3.3.11.1
Under voltage protection	3.3.11.3
Form factor	3.5.2
Earphone connection	3.5.3.3.5
Receiver radiation	3.4.3.5
Overload protection	3.3.11.2

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Environmental tests (In accordance with the indicated procedures of MIL-E-5272 except where otherwise specified herein)

Test	Paragraph	Procedure
Vibration	3.3.8.1	XII curve A (Operation is not required during vibration) Paragraph 4.7.12.3 of MIL-E-5272 shall not apply.
High temperature	3.3.8.2	II except that high temperature shall be 60°C
Low temperature	3.3.8.2	I except that low temperature shall be -30°C
Altitude (beacon operation)	3.3.8.3	VI except that temperature shall be -30°C and pressure shall be 5.54 inches Hg (40,000 feet)
Altitude (voice and receiver)	3.3.8.3	VI condition B except that temperature shall be -30°C
Explosive decompression	3.3.8.3	Contractor procedure
Altitude equalization	3.3.8.3.1	Contractor procedure including speaker and microphone operation
Sand & Dust	3.3.8.4	I except that part II high temperature shall be 60°C
Fungus	3.3.8.5	I (Operation is required at completion of test)
Immersion	3.5.3	Contractor procedure
Humidity	3.3.8	I except that maximum temperature shall be 60°C
Overvoltage	3.3.11.5	Contractor procedure

4.2.3 Preproduction approval. Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

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4.2.4 Production equipment. Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction sample. Evidence of noncompliance with the above shall constitute cause for rejection, and for equipment already accepted by the Government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

* 4.3 Preproduction reliability tests. The contractor shall conduct a preproduction reliability test in accordance with MIL-STD-781 using Test Plan XXVI Test Level E (except that failures occurring during the heating cycle prior to reaching -30°C which correct themselves later shall be considered nonrelevant). Failures occurring prior to reaching -30°C which remain afterward shall be considered relevant. Sample size shall be consistent with the requirements of Test Plan XXVI.

4.4 Acceptance tests. The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance of approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

a. Individual tests

b. Sampling tests

c. Reliability assurance tests

d. Special tests

4.4.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have to pass the following tests:

a. Examination of product

b. Operational tests

c. Manufacturing burn-in test

4.4.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met.

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4.4.1.2 Operational test. Each equipment shall be operated to check sufficient characteristics and adequate data shall be recorded to assure satisfactory equipment operation. This test shall by performed in accordance with the approved test procedure of 4.5 and shall include the following:

Parameter	Paragraph Reference
Transmitter power	3.4.2.1
Transmitter frequency	3.4.2.2
Modulation capability	3.4.2.3.2
Beacon modulation	3.4.2.4.2
MCW modulation	3.4.2.4.2
Side tone	3.4.2.4.3
Receiver sensitivity	3.4.3.1
Audio output level	3.4.3.3.1
Volume control	3.4.3 3.2
Input current	3.3.11.1

* 4.4.1.3 Burn-in period. The contractor shall operate all qualification and production units for a period of twelve hours prior to intiation of reliability or performance tests. Failures occurring during this period may be considered nonrelevant, but shall be recorded and reported. During this period, equipment shall be at room ambient temperature and humidity, and shall be vibrated for ten minutes of each hour at 2.2 G (2156 cm/sec^2) ±10 percent peak acceleration at any frequency between 20 and 60 Hz. Operation time shall be divided equally among the six modes of operation (243 MHz beacon, mcw, voice transmit and receive, and 282.8 MHz voice transmit and receive). Additional burn-in may be conducted at the contractor's option.

* 4.4.2 Sampling tests. Equipments selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling for sampling tests by the Government inspector in accordance with the following: (Sampling tests shall be conducted only when Reliability Assurance Tests of 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

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* 4.4.2.1 Scope of tests. As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational test (per 4.4.1.2) at ambient room temperature conditions, making and recording all measurements required to assure that all applicable specification requirements have been met.

b. Operational tests at selected environmental conditions. Conditions shall be selected from those of 4.2.2 and shall be based on results of preproduction, initial production, individual and special tests.

* 4.4.3 Reliability assurance tests. The contractor shall conduct monthly reliability assurance (production verification) tests in accordance with MIL-STD-781 using Test Plan III Test Level E except that failures occurring during the heating cycle prior to reaching -30°C which correct themselves afterward shall be considered nonrelevant. (Failures occurring prior to reaching -30°C which remain afterward shall be considered relevant.) Sample size shall be consistent with the requirements of Test Plan III.

4.4.4 Special tests. Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment. Equipment selected for special tests may be selected from items previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special test schedule. Selection of equipments for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of tests. Special tests shall consist of such tests as are approved by the procuring activity. Test procedures previously approved for the preproduction tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

4.4.5 Equipment failure. Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

d. Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

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* 4.5 Test procedures. The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the Government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests, when deemed necessary.

* 4.6 Reconditioning of tested equipment. Equipment which has been subjected to initial production test, acceptance and life tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.7 Presubmission testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

* 4.8 Rejection and retest. Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government inspector.

* 4.9 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PACKAGING

5.1 General. All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with MIL-E-17555 and MIL-STD-794. In the event the equipment is not covered in MIL-E-17555, the method of preservation for Level A shall be determined in accordance with the selection chart in Appendix D of MIL-STD-794.

6. NOTES

6.1 Intended use. The equipment shall be used to locate and communicate with downed aircrewmen in tactical operations.

* 6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of packaging and packing (see 5.1).

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* 6.3 Test values. Normal and limiting values of performance data shall be determined at input voltages as specified herein. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standard of performance.

* 6.4 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary specification are in conflict, the following precedence shall apply:

a. Contract - The contract shall have precedence over any specification.

b. This specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

c. Referenced specification - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

* 6.5 Conditions of test performance. During performance of preproduction tests, initial production tests, acceptance tests and life tests, it shall not be necessary that the RF portions of the equipment be terminated by the flexible antenna nor that the equipment be powered by the specified battery. As a condition of acceptance, a performance confidence check shall be performed on each equipment complete with its antenna. During this confidence check, the equipment shall be cycled through all of its mode of operation to demonstrate proper operation. Parametric measurement of performance shall be made on relative measurement basis.

* 6.6 International interest. Certain provisions of this specification are the subject of international standardization agreement STANAG 3281 SAR (1st Draft-Edition No. 4), Personal Locator Beacons. When amendments, revision, or cancellation of this specification is proposed which affects or violates the international agreement concerned, the preparing activity will take appropriate reconcilation action through international standardization channels including departmental standardization offices if required.

* 6.7 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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